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Demonstration of the E-BEPU methodology for SL-LOCA in a Gen-III PWR reactor

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ABSTRACT

The paper presents the first practical application of the alternative Extended Best Estimate Plus Uncertainty (E-BEPU) methodology. The E-BEPU is the systematic risk-informed combined deterministic and probabilistic methodology dedicated to design verification of Nuclear Power Plants. The approach covers applying the best-estimate computer code, realistic input data for initial and boundary conditions with uncertainties, and plant systems availability based on Probabilistic Safety Analysis methods. Furthermore, it extends the scope of the uncertainty analysis to include the availability of safety systems as an additional uncertain item. The approach is characterized by an enhanced Defence-in-Depth; it can detect cliff-edge effects and include risk insights in the deterministic analysis framework. It is an advanced approach to Deterministic Safety Analysis beyond the BEPU approaches and an alternative realization of the Extended-BEPU. In this study, the generic Generation-III Pressurized Water Reactor design was investigated to demonstrate the methodology with a practical example. The Postulated Initiating Event was a surge line double-ended break (SL-LOCA), a variant of the Large Break Loss of Coolant Accident (LB-LOCA), and considered an example of the Design Basis Event. The considered reactor design passed successfully through the procedure with a large margin due to the robustness of its safety systems.

1. Introduction

The most fundamental reference for the Deterministic Safety Analysis (DSA) of NPPs is the IAEA Safety Standards Deterministic Safety Analysis Specific Safety Guide-2 (SSG-2). The SSG-2 Revision-0 was published in 2009 [1], and Revision-1 was published in 2019 [2], and these documents define four options for performing DSA. Following the chronological order, the first approach is Option-1, which covers a simple conservative analysis with conservative assumptions, data, and computer codes. It was introduced early in the history of Nuclear Power. Option-2 covers the usage of the Best Estimate (BE) computer code with conservative data and assumptions, and it emerged when more sophisticated computational tools became available, and the reduction of conservatism was justified. The Best Estimate Plus Uncertainty (BEPU) methodology is Option-3, which is also the current state-of-the-art in the DSA analysis of nuclear reactors [3–6]. It applies conservative assumptions on the availability of the safety systems, best estimate codes, and typically the best possible initial and boundary conditions (I&BC). Realistic I&BC conditions with uncertainty quantification are studied when proper probabilistic distributions or uncertainties are known and the conservative values in the case of insufficient knowledge. The BEPU approach was extensively developed, tested, and demonstrated for more than a quarter of a century. It was accepted as a licensing approach for deterministic assessments of reactor designs in many countries. In spite of that, the methodology is continuously studied by several research groups [3–5, 7–17].

For both revisions of the SSG-2 Guide, there are no differences between Option-1 and Option-2 and no substantial differences for Option-3. There are only very slight changes in Option-3, as can be observed in Table 1. A lack of changes indicates that the first three options can be considered developed and mature concepts. On the contrary, there are differences in Option-4, and the basic idea was updated in the SSG-2 Revision-1 as compared in Table 1.

It should be highlighted that the Option-4 in Revision-1 is a different concept than presented in Revision-0 (see Table 1). The topic is elaborated due to the importance for the E-BEPU studied in this work. In the SSG-2 Revision-0, Option-4 is an extension of BEPU (Option-3) with a

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Nomenclature

ACCU	Accumulator
AOO	Anticipated Operational Occurrences
BEPU	Best Estimate Plus Uncertainty
CCF	Common Cause Failure
CL	Cold leg
DB	Design Basis
DBA	Design Basis Accident
DBC	Design Basis Condition
DBE	Design Basis Event
DEC	Design Extension Conditions
DiD	Defence-in-Depth
DSA	Deterministic Safety Analysis
E-BEPU	Extended Best Estimate Plus Uncertainty
ET	Event Tree
FOM	Figure-of-Merit
FT	Fault Tree
HL	Hot leg
I&BC	Initial and boundary conditions
I&C	Instrumentation and Control
IAEA	International Atomic Energy Agency
ITL	Increased Tolerance Level
LB-LOCA	Large Break Loss of Coolant Accident
LOCA	Loss of Coolant Accident
LOFW	Loss of Feed-Water
LHSI	Low Head Safety Injection
MGL	Multiple Greek Letter
MHSI	Medium Head Safety Injection
MPCT/ PCT	Maximum peak cladding temperature
MPET	Multiple Path Event Tree
PC	Plant Condition
PIE	Postulated Initiating Event
PSA	Probabilistic Safety Analysis
PS	Plant State
PWR	Pressurized Water Reactor
PRZ	Pressurizer
RAC	Regulatory Acceptance Criterion
RPV	Reactor Pressure Vessel
NPP	Nuclear Power Plant
SAMG	Severe Accident Management Guidelines
SB-LOCA	Small Break Loss of Coolant Accident
SFC	Single Failure Criterion
SG	Steam generator
SL-LOCA	Surge Line Loss of Coolant Accident
SIS	Safety Injection System
SRS	Simple Random Samples
SSG	Specific Safety Guide
STL	Standard Tolerance Level
TH	Thermal hydraulics
WP	Work Package

Table 1
Comparison of options 3 and 4 for performing DSA. Tables were taken from [1] and [2].

Option	Computer code type	Availability of systems (and assumptions)	Type of initial and boundary conditions
2009 SSG-2 Rev-0 [1]			
Option 3 - Best estimate	Best estimate	Conservative assumptions	Realistic plus uncertainty; partly most unfavorable conditions (a)
Option 4 - Risk informed	Best estimate	Derived from probabilistic safety analysis	Realistic input data with uncertainties (a)
(a) Realistic input data are used only if the uncertainties or their probabilistic distributions are known. For those parameters whose uncertainties are not quantifiable with a high level of confidence, conservative values should be used.			
2019 SSG-2 Rev-1 [2]			
Option 3 - Best estimate plus uncertainty	Best estimate	Conservative	Best estimate Partly most unfavourable conditions
Option 4- Realistic (a)	Best estimate	Best estimate	Best estimate
(a) For simplicity, the terms' realistic approach' or 'realistic analysis' are used in this Safety Guide to mean best estimate analysis without quantification of uncertainties.			

reduction of the conservatism and no limitation set on the transient type. Contrary to the BEPU, the availability of the safety systems is derived from probabilistic analyses (PSA type). In effect, the whole option is referred to as a risk-informed approach with realistic input data and realistic quantification of uncertainties, when possible, and conservative when no proper data is available [1]. The conservatism reduction process between Options 3-to-4 is analogous to reduction between Options 1-to-2 and 2-to-3. The BEPU approach applies conservative assumptions on the availability of systems, and Option-4 removes it by introducing proper synergy with probabilistic methods necessary to maintain compliance with the Defence-in-Depth philosophy.

On the contrary, in the new revision of the SSG-2, Option-4 is defined in more general terms. It is defined as a realistic approach or realistic analysis, with best estimate computer codes, best estimate assumptions, and best estimate initial and boundary conditions. There is no explicit demand for the quantification of uncertainties. In the Revision-1, Options 1-3 are 'conservative analysis', where Option-4 is considered a 'realistic analysis'. What is essential, it is explicitly limited to Anticipated Operational Occurrences (AOO) and Design Extension Conditions (DEC), but Design Basis Accidents (DBAs) are not covered [2]. Therefore, it can be considered a separate option from the other three options, which apply to non-DBAs only.

What is important, the Option-4 cannot be considered mature, and it is an area of active research. Alternative approaches to Option-4 have been under development since the first publication of the SSG-2 guide. Both revisions provide only a limited number of details on how to execute it. It is worth mentioning that the E-BEPU or equivalent methodology is not explicitly mentioned and not defined as an approach to Option-4 in any IAEA official document.

The E-BEPU methodology presented in this work was developed during the Horizon 2020 NARSIS Project [18], and efforts are presented in publicly available reports [19–24]. In the WP3 (Work Package) and WP5, theoretical research was conducted, and in the WP4 practical application was prepared. A detailed discussion of the current state of the concept, ideas, philosophy, and each step of the procedure, are presented in the E-BEPU theoretical basis report D3.8 [19]. The discussion of the E-BEPU in the context of the Defence-in-Depth is presented in the report D3.9 [20]. Theoretical effort to extend the proposed methodology to DEC is considered in the D3.10 report [21]. The E-BEPU in the context of the SAMG is discussed in the D5.5 report [23]. Outcomes of the practical applications are discussed in the D4.5 report [24], and preliminary results were also presented in the conference paper [22].

The primary motivation of this paper is to share the WP4 outcomes with the community, basically the demonstration of the new E-BEPU with a practical NPP example. Additional motivation was to introduce

the considered E-BEPU and report its development status.

As mentioned previously, the methodology used in this work is a possible realization of the Option-4 presented in the IAEA SSG-2 Rev-0, but it does not fully correspond to the Option-4 in the SSG-2 Rev-1. The methodology was developed without limitation put on the transient type and with an intention to study the whole reactor design. Therefore, strictly speaking, it cannot be understood as Option-4 in the current IAEA definition. However, when applied to DEC or AOO, it can be considered a realization of Option-4. Consequently, the presented methodology is recognized as an approach to Extended-BEPU and an

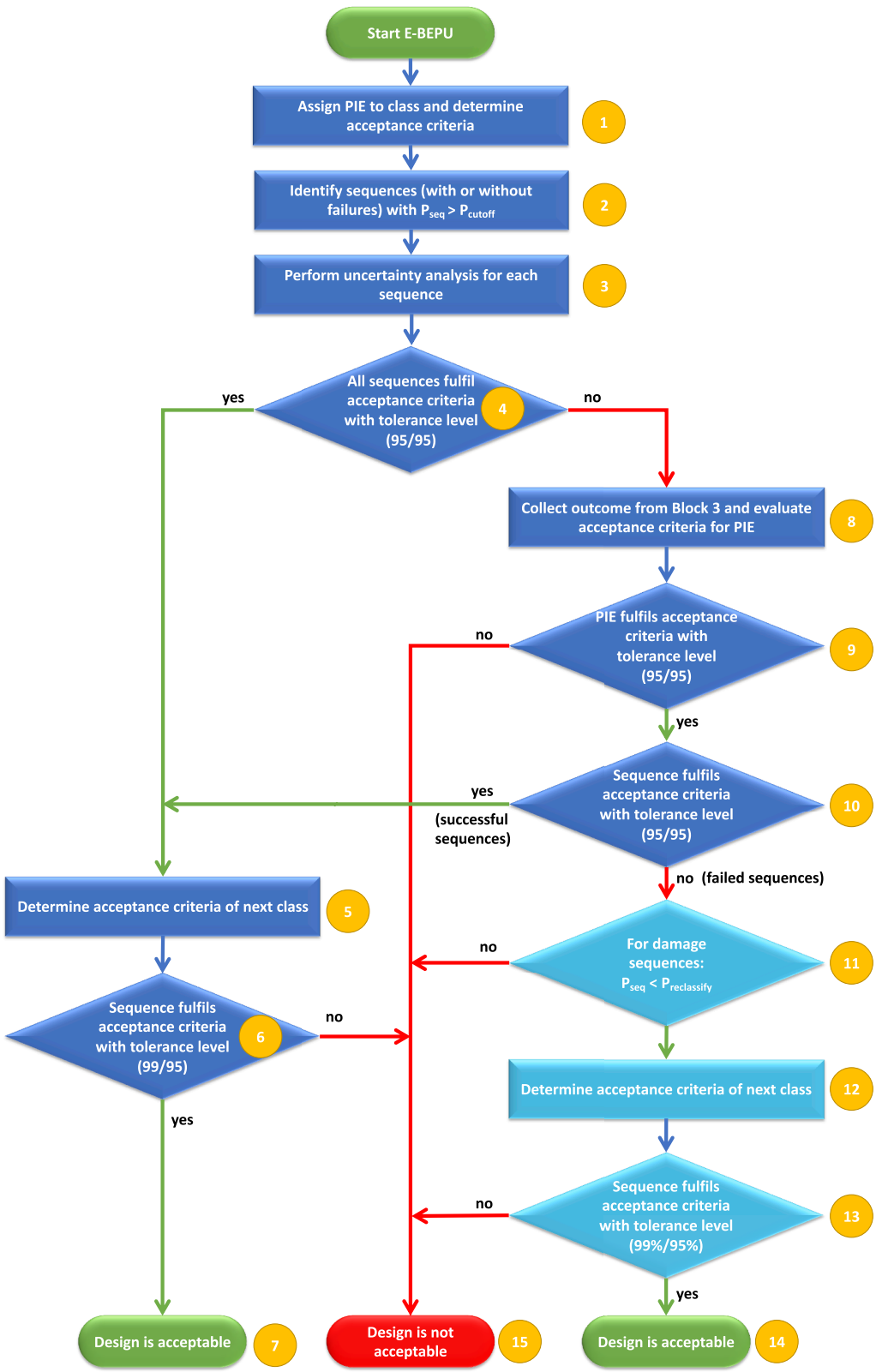

Fig. 1. Flow Diagram of the basic E-BEPU Methodology. Based on [19].

alternative approach to the risk-informed DSA.

The field of combined deterministic and probabilistic methods for the assessment of NPPs is the active research area. It is related to topics of risk-informed methods, combined or integrated methods, dynamic probabilistic methods, extended deterministic methods, and uncertainty analysis methods, including also BEPU methods. These topics are studied in the recent nuclear-related literature with various applications [8, 25–31].

There are some research papers related to the possible realization of Option-4 or extensions of the BEPU methodology. The E-BEPU methodology applied in this work is an advancement of the ideas by Dusic et al. in a 2014 paper [32] and was initially proposed as early as 2011 [33–36]. The approach was developed as a variant of Option-4, which was defined in the SSG-2 Rev-0. The alternative E-BEPU was proposed by Martorell et al. in 2017 and published in two papers [37, 38]. They applied it to study the Loss of Feedwater (LOFW) accident. Another approach that is a variant of the E-BEPU was proposed by Queral et al. in 2020 [39], where they studied the LB-LOCA accident. An alternative combined deterministic-probabilistic approach was applied by Kang et al. in 2013 [40] to study the beyond design-basis LB-LOCA for APR1400. An alternative approach called Advanced Probability Safety Margin (APSM), which can be considered an E-BEPU-like approach, was proposed in 2021 by Sun et al. [41] and applied to study SB-LOCA.

The paper is organized as follows. Section 2 is a brief overview of the methodology, and Section 3 describes prerequisites for the analysis, like models and assumptions. Section 4 presents a detailed demonstration with a discussion of each step of the methodology with a practical NPP case study. Final conclusions are presented in Section 5.

2. E-BEPU methodology overview

The E-BEPU methodology is depicted in Fig. 1. It covers fifteen steps (blocks), each representing a set of complex activities. In this subsection, the methodology is briefly presented. Then, in the following sub-sections, subsequent steps will be described in detail, and the context of the studied example demonstration case.

The whole procedure starts with Block #1 (Fig. 1); the studied Postulated Initiating Event (PIE) is classified in terms of a Plant Condition (PC), and a proper set of Regulatory Acceptance Criterions (RAC) is defined. Then Block #2 has the purpose of selecting sequences to be analysed (also called the screening process). Block #3 is similar to the typical BEPU, and it covers uncertainty analysis for each selected sequence separately. The BEPU results are assessed in Block#4. It is a point splitting the workflow into two different paths.

Blocks #4 to #7 form a path for the situation when uncertainty analysis shows that all sequences selected in Block #3 fulfill the RAC with a proper tolerance level (Standard Tolerance Level (STL), e.g. 95/95). Inside Block #5, the RAC criteria for the next accident class are defined, and inside Block #6, it is tested with a more stringent tolerance level - Increased Tolerance Level (ITL) (e.g. 99/95). If this test is positive, the design is acceptable (Block #7). This approach is a special feature of the E-BEPU, which allows the detection of possible cliff-edge effects and it enhances DiD.

The right path in Fig. 1, starting with Block #4 and ending with Block #14, is dedicated to a situation when there is at least one screened-in sequence failing (failed sequence) to fulfill RAC with STL in Block#4. In this case, the PIE as a whole demands an additional verification procedure by uncertainty analysis. In the next step, Block #8 is studied, and the case is sampled from the set of results obtained in Block #3. The number of sampled cases is selected to guarantee a proper tolerance level (STL). The conditional probability during the sampling is the same as the probabilities of sequence concerning PIE. A PIE is a set of different sequences, and it has to fulfill RAC with STL in Block #9. If the test in Block #9 fails, the design is not acceptable. In case of success, each sequence is assessed separately with additional criteria.

Sequences that succeeded in the Block #10 are transferred to the left branch, and Blocks #5 and #6 are executed similarly for the left branch. Sequences that do not fulfill RAC with STL are reclassified in Block #11. Reclassification means moving a sequence to the next plant condition category (class), and it works for sequences with the conditional probability being below a selected limit. If the test in Block #11 is a success, a sequence is tested with ITL for the next class RAC (Blocks #12 and #13), similarly to Block#6. Success ensures design acceptance; otherwise, if the ITL or reclassification tests fail, the design is unacceptable (Block#11 and Block#13).

3. Materials, methods, and initial activities

3.1. Studied NPP and deterministic model

The Generation-III Nuclear Power Plant (NPP) with a large Pressurized Water Reactor (PWR) was studied. The design is considered a generic plant, defined in the framework of the NARSIS project [42]. The plant has four coolant loops, four cold legs (CL), four hot legs (HL), four steam generators (SG), a single pressurizer (PRZ), and thermal power equal to 4500 MWth. The NPP is equipped with three basic safety injection systems important for reaching a controlled state for the considered PIE. Each train has Low Head Safety Injection (LHSI), Medium Head Safety Injection (MHSI), and an accumulator (ACCU).

Before applying the methodology, it was important to develop a deterministic NPP model being able to predict plant response. Therefore, a thermal-hydraulic system code model was developed with RELAP5 Mod3.3 [43]. The plant model nodalisation is presented in Fig. 2. It covers the reactor core, Reactor Pressure Vessel (RPV), primary coolant system, and secondary coolant system. Both passive and active safety systems and proper control systems are modeled, and all SISs are connected to the cold leg.

The active core model is divided into two cooling channels modeled by PIPE components (Fig. 2). The average main channel (component #535) covers most of the core, and there is also an additional hot assembly channel (#537). Each channel has two heat structures which model average and hot rod. Normal heat structures have a typical (average) power profile, whereas hot rods and hot assemblies have a peaked profile, as presented in Fig. 3.

Systems dedicated to suppressing Anticipated Transient without SCRAM (ATWS) and systems for long-term cooling, for the sake of simplicity and purpose of the demonstration, were not considered in both RELAP5 and PSA models (described in the next section). Furthermore, due to the same reason, the neutron kinetics model was turned off in RELAP5.

3.2. Selection of PIE and reference PSA model

The identification of a Postulated Initiating Event (PIE) is not a part of the methodology, and it has to be performed before the procedure. In this study, a Double Ended Guillotine Break of the Surge Line (SL) was selected. It is one of the scenarios postulated in the PSA model developed during the project [42]. What is important, this particular event is considered a Design Basis Accident for the studied design. Therefore, it is a Large Break Loss of Coolant Accident (LB-LOCA) or, more specifically, Surge Line LOCA (SL-LOCA). The surge line is a pipe connecting the hot leg of the Reactor Coolant System (RCS) with the pressurizer. It is assumed that the break occurs at the connection between the hot leg and surge line in the cooling loop #4 (see Fig. 2). The PIE is studied at the nominal full-power condition, and its frequency is equal to $1.3 \cdot 10^{-6}$ [1/year], assumed as a generic value based on the NUREG-6928 database [44].

The starting point for this work was the PSA Level 1 model developed with RiskSpectrum by project partners for referential NPP design [42]. The modified model was developed with RiskSpectrum and a supporting model with Sapphire software [45]. The motivation to use the existing model was to reduce effort and demonstrate the possibility of using

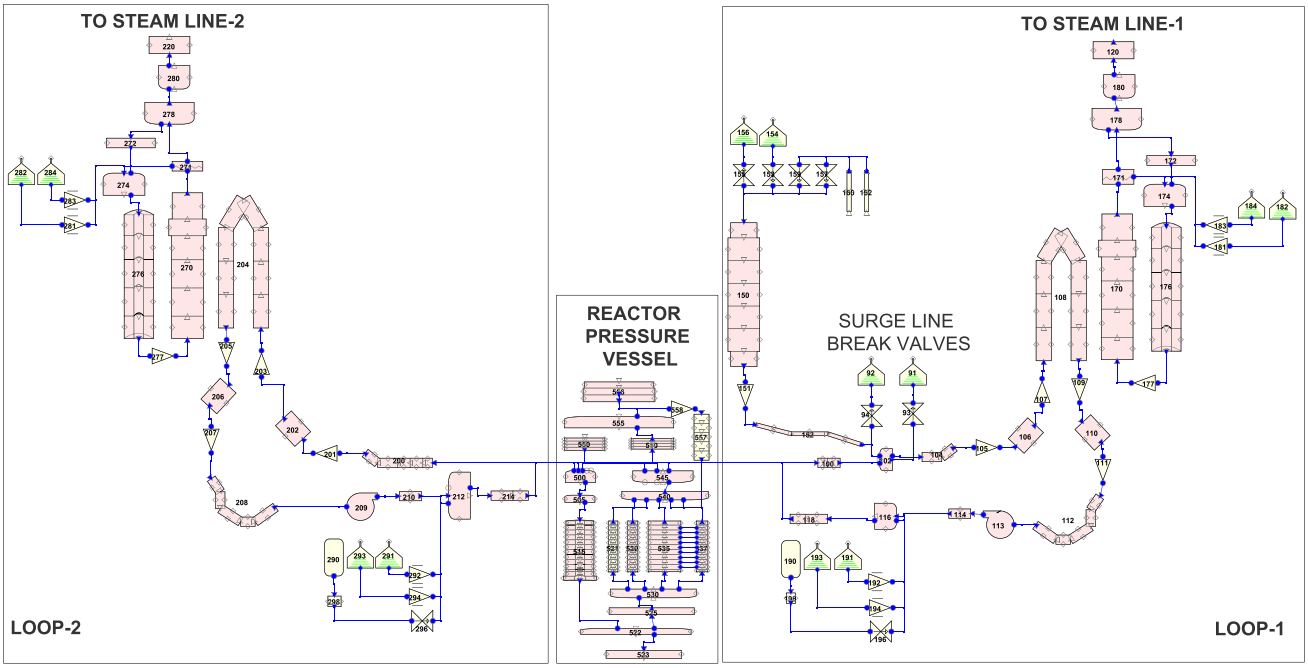

Fig. 2. RELAP5 NPP model. Loops 1 and 2 (out of 4).

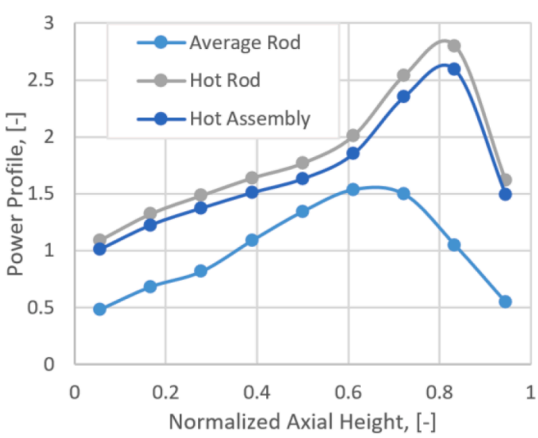

Fig. 3. Normalized power profile as a function of the active core height.

existing PSA models for the E-BEPU application. It is quite significant from a practical point of view, as E-BEPU is effort-intensive, and it is suitable to use the existing resources as far as possible.

The event tree (ET) for the standard PSA Level-1 for SL-LOCA is presented in Fig. 4. The considered ET covers only five final states, two successful and three with core damage. The whole machinery of different Fault Tree models is present in the model for all important systems. The MGL (Multiple Greek Letters) model was used for the Common Cause Failure (CCF). What should be mentioned, the PSA model also covers long-term cooling with the sixth top event, but for this study and simplicity, it was not investigated. In the ET, it adds a single additional branch for sequence No. 2 in Fig. 3. In the original PSA model, one SIS train is unavailable due to repair or preventive maintenance. Hence, in the analysis, we consider only three LHSI, MHSI, and ACCU trains to follow the approach applied in the original model. It is a 'conservative' PSA model, and it propagates in this study.

The sequence of events for LB-LOCA, represented by Top-Events of the ET (see Fig. 4, is shortly described below. After the initial break, the Reactor Protection System (RPS) is activated, and a reactor trip occurs. The probability of RPS failure and consequential ATWS is assumed to be 10^{-4} [1/year], which is a generic value. Afterward, thanks to the pressure reduction in the primary system, the MHSI safety injection starts providing water to the cold leg. In order to have success, in terms of PSA, at least one train has to work. The conditional probability of SIS failure is $\sim 6{\cdot}10^{-5}$. Due to further pressure decrease, the passive accumulators start to provide water to the system, and in order to have a successful scenario, two operating accumulators are necessary. Failure conditional probability to guarantee that is $\sim 1{\cdot}10^{-4}$. In the case of the MHSI system and successful accumulator operation, the success in terms of a controlled reactor state is expected, and sequences 1 and 2 are

LB-LOCA	Reactor Trip	1/3 MHSI available	2/3 ACCU injection	1/3 LHSI available for injection and heat removal	No.	Freq.	Cond. Prob.	Conseq.	Code
#LLOCA_AB	CRDM	SISM01	SISA02	SISL39					
					1	1.30E-06	1.00E+00	S	
					2	2.51E-10	1.93E-04	S	SISL39
					3	1.41E-10	1.08E-04	F	SISA02
					4	7.41E-11	5.70E-05	F	SISM01
					5	1.30E-10	1.00E-04	ATWS, F	CRDM

Fig. 4. LB-LOCA Event Tree for standard Level 1 PSA Model. Based on [42]. S – Success, F – Failure.

successful. The sequence with LHSI failure can lead to core damage sequence but only in the long term, and it is beyond the scope of this study. The modifications and differences between PSA traditional approach and the E-BEPU PSA-type approach will also be described in the next chapter.

The LHSI system is not crucial to obtaining the controlled state (in the scope of this work); hence it could have been removed from the analysis. Otherwise, the success in PSA does not guarantee success in E-BEPU, and the LHSI system can play a substantial role in possible scenarios. In consequence, it was not removed from the analysis.

4. Demonstration of the E-BEPU methodology with discussion

4.1. Block 1 - PIE classification and acceptance criteria

Block#1 (Fig. 1) is initiated by selecting the proper Plant State (PS) classification. The PS classes considered in this work are based on the IAEA Design Basis Condition (DBC) and Design Extension Condition (DEC) classification with 'damage' increasing and frequency decreasing with class number:

- DBC-1, normal operation
- DBC-2, anticipated operational occurrences
- DBC-3, design basis accidents with frequency 10^{-2} - 10^{-4} [1/year]
- DBC-4, design basis accidents with frequency 10^{-4} - 10^{-6} [1/year]
- DEC-A and DEC-B, design extension conditions, non-design basis

It can be observed that the PIE considered in this work belongs to the last design basis class, DBC-4.

At Block #1, it is necessary to determine proper Regulatory Acceptance Criteria (RAC). In the case of LB-LOCA and, in general, DBC events, different options are possible, and they are extensively discussed in the CSNI SMAP Report [46]. Usually, safety criteria are expressed in doses equivalent limits. However, for LOCA analysis, they are expressed in terms of decoupling criteria, i.e. peak clad temperature, maximum oxidation fraction, hydrogen generation and possibly others. An acceptance is guaranteed when fulfilling a set of these criteria. Otherwise, this work is a demonstration and single, and the simplest criterion for LOCA was selected - the maximum peak cladding temperature (MPCT or PCT) in the hottest channel cannot exceed 1204°C (1477.6K). Therefore, peak cladding temperature is considered a Figure-of-Merit (FOM) in this work.

What is more, in this analysis, we demand the achievement of the controlled state and not the safe shutdown state, as it is usually required to meet additional conditions. It is a simplification, and readers should be aware of that. The controlled state is defined, in the PSA model, as follows [42]:

- The reactor core is subcritical,
- Reactor coolant inventory is stable, or it is increasing,
- Core power is removed by coolant via the break or/and with steam generators.

The first condition is achieved for the studied case thanks to the low conditional probability of ATWS $\sim 10^{-4}$, and it is screened out in further analysis. In effect, the neutron kinetics model is turned off. Therefore, the TH analysis is focused mainly on the second and third conditions. It is worth mentioning that for the studied plant to reach safe shutdown, proper operator actions are eventually needed about 30 minutes after the accident initiation, and it is also beyond the case studied in this work.

4.2. Block 2 - sequences identification and screening

4.2.1. Sequences identification and multiple path event tree

In Block #2, we have to identify which sequences have to be analyzed at later stages of the methodology. The first part of this effort is preparing and applying a probabilistic model, which will allow sequences identification.

The E-BEPU allows to use of PSA-type models, but the analytical approach has to be different because the probabilistic part has a different purpose than a typical PSA. We are not studying the success or failure of a particular sequence, but we focus on the whole PIE. A studied PIE has to be successful, as a whole, for the design to be accepted. It is one of the main differences between the E-BEPU probabilistic approach and traditional PSA.

In order to cope with this issue, the Multiple-Path Event Tree (MPET) approach was proposed. It is based on event trees, but MPETs have multiple branches at each branching point. Thus, an MPET allows the estimation of the conditional probabilities for each sequence with different systems availability. The Fault Trees (FT) model, similar to traditional PSA, was used to obtain proper configurations of systems to be quantified in each branching vertex. Basically, the fault tree model was applied to estimate conditional probabilities for each configuration of systems. The MPET approach is equivalent to the so-called Expanded Event Trees which was studied in [39], branch event trees considered in [41], and MPET considered in [47].

In general, the model can be created from scratch in a similar fashion as the standard PSA model, but it is also possible to start with existing ETs, FTs, and the whole PSA model - to avoid excess effort - as it was applied in this work.

The MPET developed for the demonstration case is presented in Fig. 5. The model has the same top events, but it is more complex than ET in Fig. 4. There is more than one path for each branching point, and in fact, as many paths as significant safety system trains and no train state. The same fault trees were applied with the same Common Cause Failure (CCF) Multiple Greek Letter (MGL) model as in the original PSA model. In order to develop MPET, it was necessary to recalculate frequencies for considered sequences, and the final conditional probability values for each sequence are depicted in Fig. 5.

What is an important topic for the E-BEPU is that outcomes based on the normal PSA model cannot be enough for the analysis. The MPET (Fig. 5) has the same top events as PSA, but it is not expected to be a general rule for the E-BEPU. Analysts have to use deterministic simulations to verify the ET for E-BEPU and eventually develop additional top events in ET. Proper verification is necessary and was also the case for LB-LOCA studied in this work. However, results showed that it is a secondary importance problem for this particular PIE. For other PIE, typically from lower DBC classes, we can have a substantial impact on the state of the plant and the analysis.

Worth mentioning that a large break in the primary system has such large consequences that the important effects are mainly due to the operation of safety injection systems, typically designed to cope with LOCAs.

For this work, we can identify sequences by a vector containing four numbers. It defines the state of systems involved in top events in Fig. 5. The vector prototype is S={RT, MHSI, ACCU, LHSI}. For example, some sequence vector S=1333 defines the first sequence in Fig. 5, and it means that RT is successful, three MHSI, three accumulators, and three LHSI are available. We can use the "-" sign for the sequences where systems are not important or do not analyze them. For example, in the case of the ATWS sequence, S_ATWS = {0—}.

It is also worth observing that some conditional probability values in Fig. 5 might be unexpected, as, for example, sequence 1213 has a slightly lower probability than 1212. According to the structure of the MHSI and LHSI trains with the same number of the system have common sub-systems; therefore, if N trains of MHSI fail it is more probable that the same number of LHSI trains fails.

4.2.2. Sequences screening

In the next phase, probabilistic results are used in the Block #2 to perform the sequence screening process. We are performing the

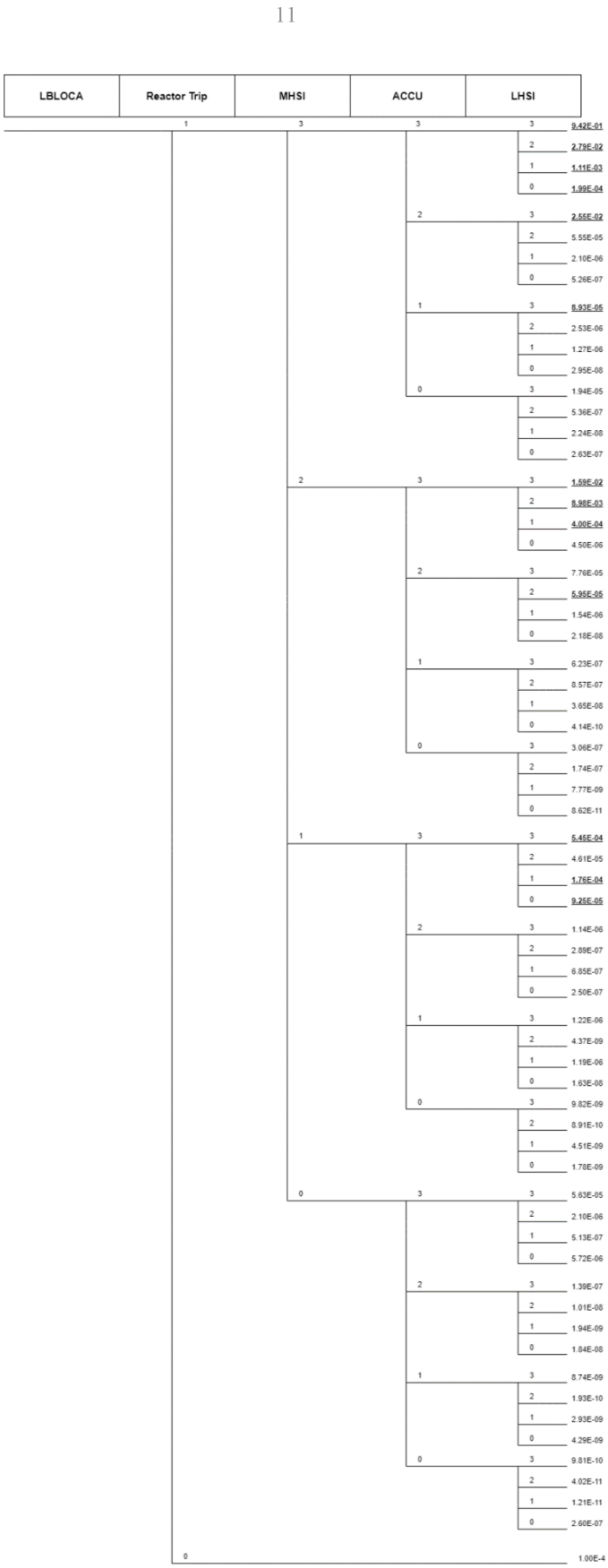

Fig. 5. Multiple Path Event Tree for studied PIE. The last row defines a conditional probability for sequences in relation to the PIE. Underlined sequences were selected for further analysis.

screening process for all sequences, no matter if the sequence is failed, leading to core damage or not. The proper cut-off (screening) criteria were proposed that the sum of all conditional probabilities for screened sequences has to be lower than a certain threshold limit.

The threshold limit equal to $5{\cdot}10^{-4}$ was assumed in this work. The value was recommended in EBEPU Manual [19] and is based on evaluating the minimum conditional probability value below which sequence can be removed from uncertainty analysis of a PIE without significant impact on tolerance levels [19]. Modification of screening criteria to other values may change the E-BEPU results, and it demands special attention.

In the screening process, there is also a second criterion. In the DSA, and especially for DBA, a proper level of conservatism has to be present to preserve the DiD (Defence-in-Depth). Typically, a PIE is assumed in parallel with a conservatively selected Single Failure Criterion (SFC). In order to ensure DiD, all sequences with a single failure cannot be screened out, no matter what their conditional probability is [19].

Finally, thirteen different sequences were identified at the end of Block #2, as presented in Table 2 and Fig. 5. These sequences were not screened out and are used in further analysis.

Sequences #1, 2, 3, 4, 5, 6, 7, 8, 9, 11 were not screened due to large probability. When sequences 1-13 are studied, the conditional probability for screened sequences is $3.84{\cdot}10^{-4}$ (below the limit of $5{\cdot}10^{-4}$). Sequence 12 has a probability of $9.25{\cdot}10^{-5}$, and initially, it was screened-out, and the total conditional probability was $4.76{\cdot}10^{-4}$. Finally, it was decided to include this sequence in the analysis. The decision is more or less arbitrarily, the sequence has a relatively large probability, and it was considered as a limiting situation showing the abilities of a single MHSI system train because LHSI is not available (sequence 1130).

What is more, sequence 13 (1222) has a low probability, but it can be considered as the Single Failure Criterion (SFC) scenario. The worst considered SFC is the failure to open the isolation check valve for SIS, which results in the loss of one MHSI, LHSI, and ACCU. Other sequences that can be considered a single failure for each SIS function are MHSI, or LHSI or ACCU are sequences 2, 5, 6. Nevertheless, these sequences have a large probability, and they should not be considered during the screening process.

4.3. Blocks 3 and 4 - uncertainty analysis

4.3.1. Uncertainty analysis for each sequence (Block 3)

Sequences that successfully passed Block #2 are studied in Blocks #3 and #4 (see Fig. 1). Each sequence is analyzed separately. Block #3 is analogous to the current BEPU practice, and in this work, we applied the approach based on the GRS methodology [48]. The procedure is depicted in Fig. 6. It is a popular approach for uncertainty propagation based on the Wilks method, which can be classified as a black-box from the point of view of modeling, and it has input-driven uncertainty

Table 2
Sequences that were not screened in Block #2. MHSI – Medium Head Safety Injection, ACCU – Accumulators, LHSI – Low Head Safety Injection.

No.	Code	MHSI	ACCU	LHSI	Conditional Prob.
1	1333	3	3	3	9.42E-01
2	1332	3	3	2	2.79E-02
3	1331	3	3	1	1.11E-03
4	1330	3	3	0	1.99E-04
5	1323	3	2	3	2.55E-03
6	1233	2	3	3	1.54E-02
7	1232	2	3	2	8.98E-03
8	1231	2	3	1	4.00E-04
9	1133	1	3	3	5.28E-04
10	1313	3	1	3	8.93E-05
11	1131	1	3	1	1.76E-04
12	1130	1	3	0	9.25E-05
13	1222	2	2	2	5.95E-05

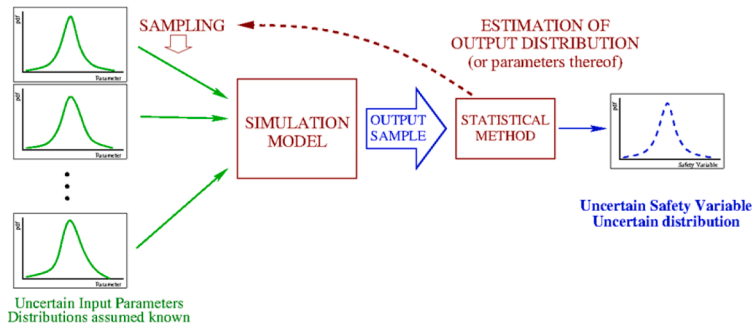

Fig. 6. BEPU approach used in this work. Taken from [48].

propagation [48, 49]. In principle, the E-BEPU methodology is neutral from the point of view of the BEPU approach [19]. Wilks theorem-based approach is likely the most popular and is considered a non-parametric frequentist type, but also parametric methods are possible. Worth mentioning that, Bayesian-type parametric and non-parametric methods are available [6, 17, 50]. The extensive literature about BEPU approach exists including several recent publications, inter alia [6, 8, 9, 17, 40, 50, 51].

The first step was selecting input parameters being significant for the safety output or Figure-of-Merit (FOM) and selecting proper probability distributions. Selected parameters with proper distributions are presented in Table 3. These are typical for LB-LOCA accidents, and they are based on the considerations presented in the literature for different BEPU LB-LOCA studies [14, 52, 53]. We assumed that the set of parameters is the same for all studied sequences, but it does not have to be true in general for other PIEs and other attempts to E-BEPU.

In the next step, a Simple Random Samples (SRS) sampling was applied for uncertain parameters. The Wilks formula provides the number of computational cases necessary to obtain proper confidence and tolerance level. In this case, the 95%/95% one-sided tolerance limit, the so-called STL (Standard Tolerance Limit), is recommended. According to Wilks, it corresponds to 59 computational cases [49]. The procedure was repeated for each sequence selected in the Block #2 screening process. It resulted in 13×59=767 system code models, and code runs necessary to complete Block #3 [48]. Then, the simulations were performed, and proper outputs were generated and post-processed. Transients were simulated for 1800 seconds because long-term plant stabilization was not considered, and all-important events occurred within this time interval.

4.3.2. Assessment of the uncertainty analysis results (Block 4)

Tolerance limits for the safety outputs (FOM) were obtained with the Wilks method. It is a part of Block #4 where all simulated sequences, one by one, are compared with acceptance criteria (RAC). If at least one sequence does not fulfill acceptance criteria, we proceed to Block #8 (Fig. 1) and the right part of the methodology. Otherwise, if all sequences fulfill the acceptance criteria, we proceed to Block #5 (Fig. 1) and the left branch of the methodology.

Fig. 7 presents maximum peak cladding temperature curves for each scenario. Maximal curves are understood as PCT curves for cases with the highest observed PCT for a particular sequence for a set of results calculated with STL (1-out-of-59).

It can be observed that RAC was not violated in any run. Therefore, there is a very large margin to limit (~737K). It shows that the studied PIE is successful from the point of view of Block #4. In consequence, the studied test case follows the left branch of the procedure (Fig. 1). The results were expected as the considered NPP design is very robust, equipped with properly designed safety systems dedicated to Loss of Coolant Accidents. What is more, the studied PIE, SL-LOCA, is less violent than hot-leg or cold-leg break. The SL-LOCA is considered a design basis event due to applying the break preclusion principle for RCS pipework. However, this principle does not have to be accepted by all Regulatory Bodies for LB-LOCA accidents. Hence, the presented outcome works for the studied design in case of validity of the Break Preclusion Principle.

4.3.3. Discussion of uncertainty analysis validity

Additionally, uncertainty results for SFC sequence 13 (1222) are presented in Fig. 8 to indicate typical uncertainty limits for this 'serious'

Table 3
Uncertainty parameters with probability distributions.

No	UncertaintyParameter	Nominal value	Unit	Type	Distribution	Min	Max	Mean	Std. Dev
1	Break discharge coefficient	1.0	—	multiplicative	normal	0.95	1.05	1	0.025
2	Initial power	4500	MWth	multiplicative	normal	0.98	1.02	1	0.01
3	Fuel thermal conductivity	correlations	—	multiplicative	normal	0.9	1.1	1	0.05
4	UO2 specific heat	correlations	—	multiplicative	normal	0.98	1.02	1	0.01
5	Fuel gap size (diametric gap)	0.17	mm	multiplicative	normal	0.8	1.2	1	0.1
6	Power after scram decay power multiplier	Maximum Decay + 2sigma	—	multiplicative	normal	0.92	1.08	1	0.04
7	I&C signal delay start pumps – MHSI, LHSI	15	s	Additive	uniform	15	40	—	—
8	Initial conditions: primary loop flow	27185	m^3/h	multiplicative	normal	0.96	1.04	1	0.04
9	Initial cold leg temperature	295.6	°C	additive	normal	-2	2	0	1
10	Accumulator initial pressure	4.5	MPa	additive	normal	-0.2	0.2	0	0.1
11	Accumulator line friction form losses	8.65	—	multiplier	log-normal	0.5	2	1.25	0.75
12	Accumulator temperature	50	°C	additive	normal	-10	10	0	10
13	MHSI - flow characteristics	table	kg/s	multiplier	normal	0.95	1.05	1	0.05
14	LHSI - flow characteristics	table	kg/s	multiplier	normal	0.95	1.05	1	0.05
15	Hot channel peaking factor	2.82	1	multiplier	normal	0.95	1.05	1	0.025

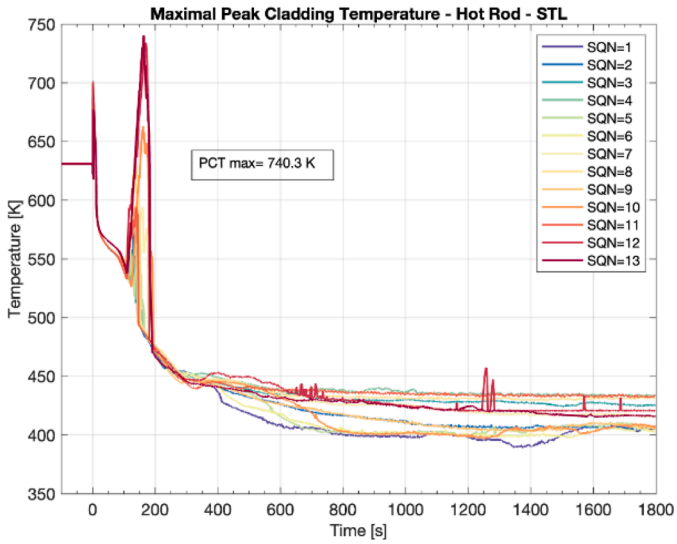

Fig. 7. Peak Cladding Temperature curves for sequence realizations characterized by Maximum Peak Cladding Temperatures for Standard Tolerance Limit (STL), obtained during uncertainty analysis. RAC Limit 1477K (not visible).

sequence in the case of average and hot rods. It is worth mentioning that this sequence could be considered for this PIE as a standard BEPU case [4, 7]. On the contrary, Fig. 9 presents uncertainty results for fully successful sequence 1 (1333).

The obtained uncertainty band can be considered narrow. The reader has to remember that the considered analysis has the main aim to demonstrate how the methodology works, and it is not an attempt to assess a real plant or perform a full-scale assessment. Applied uncertainty parameters can be a matter of change in case of more detailed analysis. Moreover, in the analysis, we have not varied code-specific modeling parameters, e.g. correlations for heat transfer. Instead, we have followed typical BEPU analyses available in the literature, e.g. BEMUSE analysis for LB-LOCA in Zion NPP [52] or [14]. In consequence, the uncertainty analysis applied in this can be considered by some researchers as more of an application of uncertainty analysis techniques for the quantification of uncertainties.

4.3.4. Discussion of NPP model validity

Typically the thermal-hydraulic model qualification is a primary importance technical enterprise in the DSA for NPPs. The reader should be aware that the considered plant is generic, and the model was developed as a demonstration tool and a part of a research effort to present the E-BEPU methodology. In consequence, the qualification process for SL-LOCA was limited to qualitative comparison with the ROSA experiment [54–56]. It is a simplistic approach, and a strict comparison would demand the application of more sophisticated approaches like the Kv-scaled method or Triad method ([57–59]). A fully systematic, complex process of model qualification has to be performed for any real design [60].

Fig. 10 presents PCT results for RELAP5 runs with a nominal modeling setup (see Table 3), compared with ROSA-2 Test 1 [55]. This experiment considered the thermal-hydraulic tests being similar to the

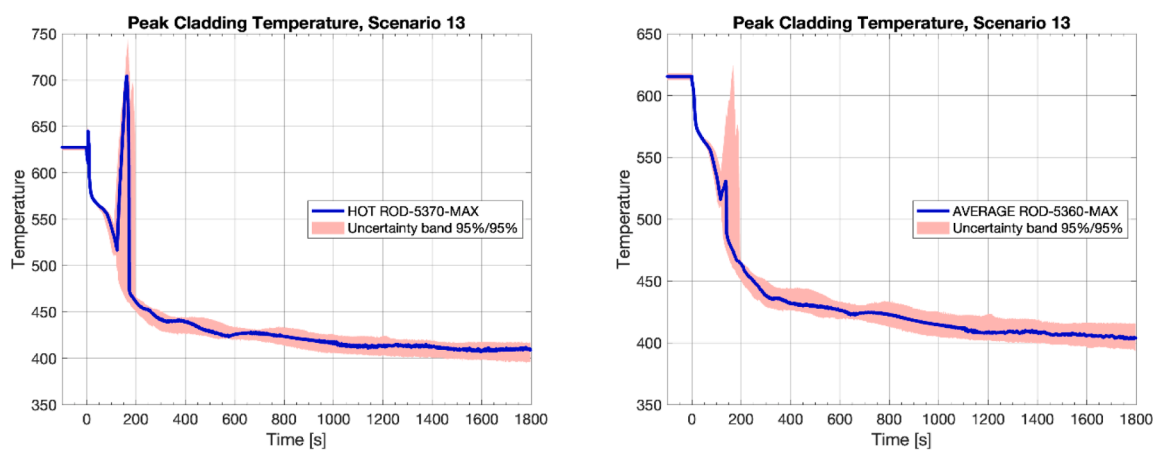

Fig. 8. Uncertainty results for sequence 13 (1222). Left – Hot Rod Results. Right – Average Rod Results.

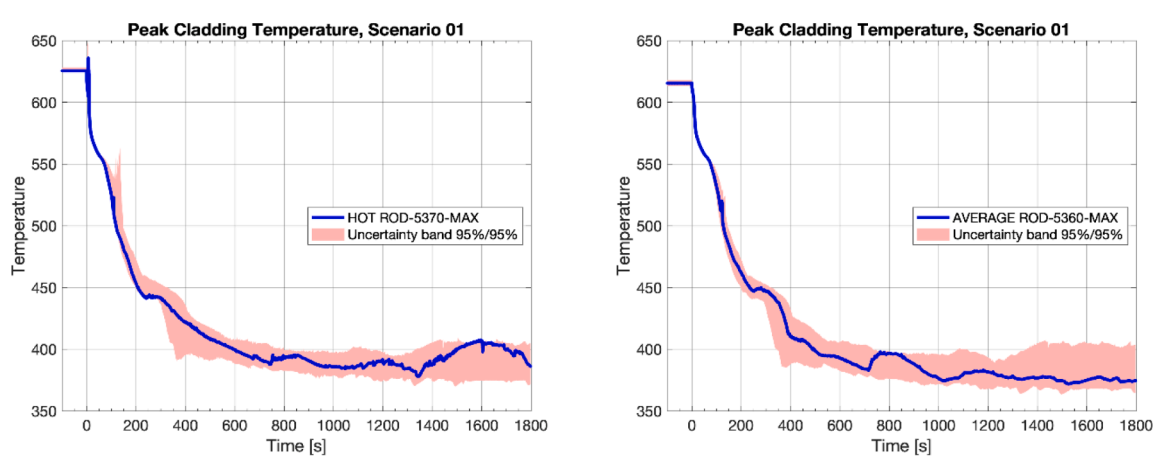

Fig. 9. Uncertainty results for sequence 1 (1333). Left – Hot Rod Results. Right – Average Rod Results.

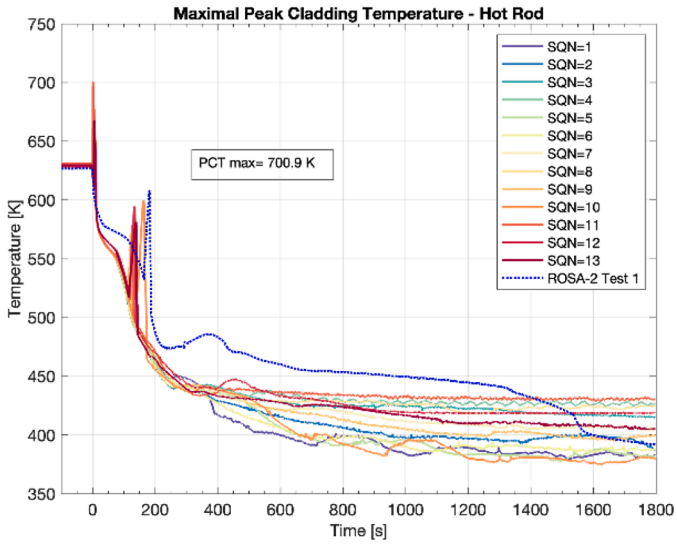

Fig. 10. Best estimate cases for STL results compared with ROSA-2 Test 1.

studied scenario. The initial temperatures were similar, break size had similar characteristics, and transient evolution was comparable. We can observe that the RELAP5 model for hot rod predicts a temperature spike at the initiating event. The spike was not present in the experiment, but we should observe that the peak is also not present for average rods, as presented in Fig. 9 (Right). Hence, the peak occurs for thermally loaded rods.

Moreover, we can observe that the secondary temperature peak occurs slightly earlier for the studied model, but the maximal difference is less than 100 seconds. The secondary peak has a comparable magnitude to the worst-case sequences. We can observe that the ROSA experiment predicted slightly higher cladding temperatures for the remaining part of the transient.

Finally, We can conclude that the applied model provides physically reasonable thermal-hydraulic results similar to the ROSA experiment. Therefore, the presented model is applicable as a tool for demonstration purposes.

4.4. Block 5 to 7 - all sequences fulfilling RAC with STL

4.4.1. Next class RAC (Block 5)

In the case of all sequences fulfilling RAC with STL in Block #4, it corresponds to the situation when PIE (as a whole) fulfills RAC [19]. In the E-BEPU methodology, it is also necessary to perform Block #5 and #6, where all sequences are tested for cliff-edge effects. In Block#5, acceptance criteria for the next (higher) class are defined.

The PIE studied in this work is a special case as it is the last design basis class (DBC-4), and above it, we have DEC conditions. Early developments presented in E-BEPU Manual proposed to omit Block#5 and Block#6 in the case of DBC-4 [19], as there are no more stringent criteria [22]. Further research indicated that it would be more appropriate not to omit Blocks #5 and #6 and include them in the DBC-4 analysis [21].

Two approaches were considered, but they are equivalent to the case studied in this work. In principle, it is not obvious which is more appropriate, and it should be a matter of more detailed research in the future. First of all, it is possible to test DBC-4 RAC with ITL. Hence we can apply the same RAC but with stronger criteria for tolerance level. Alternatively, we can treat DEC-1 (complex sequences without core melt) criteria as a next-class acceptance criterion for DBC-4 and directly follow the algorithm presented in Fig. 1. Intuitively, the first approach is more demanding, but it does not have to be a case for all PIEs.

What is also important, for recent nuclear plants design, the acceptance criteria for DEC-1 can be the same as for DBC-4. In light water reactors, the intensive exothermic reaction appears for cladding temperatures larger than RAC 1477K, and it is practically equivalent to the occurrence of serious fuel damage. Hence, it seems reasonable to consider similar decoupling criteria for DEC-1 and DBC-4, it simplifies analysis, and radiological assessment does not have to be performed. In such a situation, both alternatives for treatment in Block #5 and #6 are equivalent. It is the case in this work; the design considered has the same acceptance criteria for DBC-4 and DEC-1.

4.4.2. Uncertainty Analysis with ITL and Design Assessment (Block 6 and 7)

In Block#6, sequences are tested with Increased Tolerance Level (ITL) for compliance with the next class acceptance criteria determined in Block#5. The approach is analogous to Block #3. The uncertainty analysis is repeated with a large number of input decks. For example, for a one-sided tolerance level 99/95, the Wilks formula demands 299 input-decks [49]. If any case violates the acceptance criteria, the design is not acceptable from the point of view of the E-BEPU.

In the case of the studied design, simulations were performed separately from Block #3, but in principle, results can be shared between blocks. A total of 3887 (13×299) RELAP5 runs were executed for all studied sequences.

Fig. 11 presents maximum cladding temperatures limiting curves obtained for each sequence. Comparing with STL (Fig. 7), we can observe ~40K larger temperature peaks, especially for sequences 12 and 13. It is a non-negligible difference, and it clearly shows the impact of a larger tolerance level.

We can conclude that the outcome of ITL analysis for the studied design is the same as for the STL analysis in Block #4. The design of safety systems is so robust that it passed ITL verification successfully with a substantial margin. Consequently, the whole E-BEPU procedure for the studied design and in the case of PIE is successful, and we reach Block #7.

At this point, the demonstration is finished considering the scope of this study. However, it is important to discuss the remaining right part of the E-BEPU algorithm, and the discussion is presented in the following section.

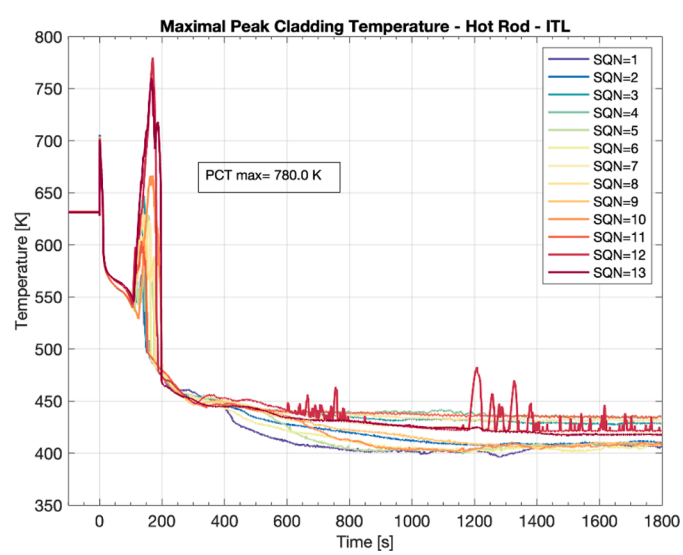

Fig. 11. Peak Cladding Temperature curves for sequence realisations characterized by Maximum Peak Cladding Temperatures for Increased Tolerance Limit (ITL), obtained during uncertainty analysis. RAC Limit 1477K (not visible).

4.5. Block 8 to 15 – not-all Sequences fulfilling RAC with STL

4.5.1. PIE uncertainty analysis (Block 8)

During the analysis for Block #4, it can happen that a PIE fails the test. If there is at least one non-screened sequence that does not fulfill RAC with STL, we proceed to Block #8 and the right branch of the methodology (Fig. 1).

As there are sequences that do not fulfill acceptance criteria with STL, we need to guarantee that the whole PIE fulfills RAC with STL. In order to perform this task, the uncertainty analysis for the whole PIE has to be performed [19].

The procedure starts by gathering uncertainty results computed with STL for all screened sequences in Block#3. We treat this dataset as a random sample of possible outcomes of PIE. In this study, it is 767 realizations. Then, we sample 59 cases to guarantee a proper tolerance level. The sampling procedure is the following. Firstly, the sequence type is randomly sampled with discrete probability distribution corresponding to the conditional probability of a sequence (Table 2). Then using uncertainty results for the sampled sequence, we sample a single realization (1-out-of-59) again with uniform probability. We assume that each realization is equally probable. The sampled realization of the sequence is added to the list of realizations that are not allowed to be sampled again (sampling without repetition). In the next step, the procedure is repeated until we obtain 59 cases corresponding to the STL for the whole PIE.

4.5.2. PIE test for RAC (Block 9)

In Block #9 (see Fig. 1), we are using the results of the PIE uncertainty results obtained in Block #8. If the whole PIE consisting of 59 sampled cases is fulfilling the RAC with 95/95 tolerance (STL), the PIE is also fulfilling the RAC with STL, and we can proceed to Block #10. The current plant design is not acceptable in the opposite case, and we transfer to Block #15 (Fig. 1).

4.5.3. Sequences test for RAC (Block 10)

A whole PIE has to fulfill the RAC in Block #9 (see Fig. 1) in order to successfully pass to the next step of the procedure. In the case of success, in Block #10, we have to use the results of Block #3 and test each sequence for RAC with STL separately. It is worth observing that in Block #4, the same test was performed, and its outcomes can be used in Block#10. Consequently, if a sequence fulfills the criteria, it is transferred to Blocks #5 and #6. On the contrary, if the sequence is not fulfilling RAC with STL, we proceed with it to Block #11.

4.5.4. Sequence reclassification (Block 11)

In Block #11, we attempt to assess the potential impact of failed (damaged) sequences on the design in the case when a PIE is fulfilling the RAC. The analysis is performed for each sequence, one by one. The studied sequences, which were not screened out in Block #3, cannot be removed from the analysis, but the E-BEPU allows moving a sequence to the next class and applies less stringent acceptance criteria. It is possible when a sequence conditional probability meets the so-called reclassification requirements.

The reclassification is introduced, and it provides a threshold value that limits the conditional probability of the sequence, which, even if failed, does not pose a substantial risk for the design. Moreover, it demands additional checks, which are performed in Block #13, in order to maintain a proper level of DiD.

The issue of selection of the reclassification limit, $P_{reclassify}$, is not an obvious topic. In this work, we follow the recommendation from [19] to estimate $P_{reclassify}$ as the ratio of lower to the upper-frequency boundary for the given PIE classification (DBC), and it is 1E-6/1E-4 = 1E-2. In consequence, if the conditional probability of the damaged sequence is larger than 1E-2, it is not possible to reclassify, Block #11 fails, and we proceed to Block #15. The design is not acceptable and demands revision and modifications. In the opposite case, we can proceed to Block #12.

4.5.5. Next Class RAC and Uncertainty Analysis with ITL (Blocks 12 and 13)

In Blocks #12 and #13, the damaged (failed) sequences are studied with conditional probabilities being low enough. First of all, in Block #12, we are determining the acceptance criteria for the next class. In principle, it can be defined the same as in Block#5.

In Block #13, due to low probability, we 'virtually' transfer the sequence to the next class, and we demand that the sequence fulfills the next class acceptance criteria with increased tolerance level ITL (99/95). This step ensures us that the sequence does not introduce serious cliff-edge effects, and we ensure a proper level of DiD.

4.5.6. Design assessment (Block 14 and 15)

If all sequences are successfully checked, the design is acceptable (Block #14) from the perspective of a studied PIE. In the opposite situation, design revision and modifications are necessary (Block #15), as was mentioned earlier.

5. Conclusions and summary

The paper presents an alternative Extended Best Estimate Plus Uncertainty (E-BEPU) methodology and its first application. The methodology was developed in the framework of the NARSIS research project, and it is an update of the earlier methodology proposed by Dusic et al. [32]. The approach was demonstrated with a design basis type event, Surge Line Large Break Loss of Coolant Accident, for generic Generation-III NPP with a PWR reactor. This accident is considered the largest LOCA in the DBA domain for the studied design due to the application of the 'Break Preclusion Principle'.

The considered design passed the E-BEPU procedure successfully for the considered design basis accident. It is directly supported by the observation that the acceptance criterion and next class acceptance criterion were not violated in any thermal-hydraulic system code run for over 3000 simulations. It is thanks that the considered NPP design is robust with quadruple redundancy for all safety injection systems with high reliability. A lot of focus is present during a typical PWR plant design process to cope with large LOCAs, and a single train is usually enough to safely shut down the plant. In the studied case, large margins to acceptance criterion were observed because studied PIE surge-line LOCA is less severe in terms of peak cladding temperature than hot-leg or cold-leg LB-LOCA, typically studied for many designs and which in this case belongs to beyond design domain. The presented outcomes show that the E-BEPU methodology applies to NPPs analysis with available DSA and PSA methods/models and can be executed with large but still reasonable computational resources.

The combined probabilistic-deterministic and risk-informed methods, like E-BEPU, can be considered the next step in the safety analysis of nuclear reactors. More research activities in this field are expected in the near future. It is because the basic objective of these methodologies is to increase the accuracy of available safety margins assessment, which can result in enhanced operational flexibility, allow power uprates, extend fuel cycles, or solve plant aging issues [19]. All these topics are becoming important with the increasing age of existing reactors and challenging economic constraints.

The application of the E-BEPU opens the possibility of decreasing the level of conservatism in the assumptions about safety systems. The current approach, like BEPU, is more conservative, and applying a risk-informed approach can be useful in deterministic safety analysis applications to include risk insight and guarantee safety despite the reduction in conservatism. The E-BEPU provides a systematic approach to studying NPP safety and allows for strengthening Defence-in-Depth. However, several features must be studied in detail, demonstrated, and stress-tested in future research.

The methodology presented in this paper can be briefly discussed

and compared with other recently developed probabilistic and deterministic combined methods equivalent or similar to E-BEPU (also mentioned in the introduction section) to indicate the most important differences and novelty [37–41]. The Extended BEPU methodologies attempt to reduce the level of conservatism in the definition of scenarios studied in the safety analysis. All mentioned approaches take advantage of PSA methods to identify possible accident sequences. In all works, with Kang et al. [40] being an exception, some types of event trees with multiple branches are considered to identify sequences.

The main feature of the E-BEPU methodology studied in this work is to include the availability of safety systems as an additional uncertain item in the uncertainty analysis. Among revised literature, this feature is also present in work by Martorell [37, 38]. The important difference to all other methodologies is that it applies the multiple-level (two-level or three-level) uncertainty analysis with a separate analysis branch (see Fig. 1) for failed cases with reclassification. The first level is the uncertainty analysis for each sequence in an extended event tree (Step 3, Fig. 1). The second level is for failed cases, where uncertainty analysis is performed for a whole PIE, so all sequences are studied simultaneously (Step 9, Fig. 1). The third level is executed for both failed and successful cases, where uncertainty analysis is performed with an increased tolerance level (Step 6 and Step 13, Fig. 1) and additional (next-class) acceptance criteria. None of the alternative E-BEPU variants apply reclassification of low probable failed sequences into the next class (Step 10, Fig. 1) [19]. Additionally, none of the alternatives use more strict criteria with ITL and additional next-class acceptance criteria. Thanks to that, the proposed E-BEPU approach strengthens DiD and opens a path to detect or avoid possible cliff-edge effects [19]. Furthermore, only this approach explicitly demands to study of Single Failure Criteria sequences even if probabilistic analysis suggests removing them. What is also important, most of the approaches in the literature follow the definition of SSG-2 Rev 1 and are dedicated to AOOs and/or DECs, whereas the presented E-BEPU was designed for all DBCs.

The basic trade-off of using E-BEPU and reducing the conservatism is the substantial increase in the necessary computational effort and complexity of the task. In basic BEPU, to study LOCA, we can perform about 59 computational cases, and in E-BEPU, the computational effort increases by about two orders of magnitude. It is an obvious drawback of the methodology, but the amount of information about an NPP safety increases, and it is possible to avoid cliff-edge effects and escalation of non-expected circumstances. It is possible to reduce the number of computational cases by the additional screening of sampled sequences on the basis of preliminary sequence progression, as was presented in work by Sun [41]. Nevertheless, this method can potentially reduce generality and lead to the unintentional omission of sequences with cliff-edge effects and remove the benefit of using a more complex approach.

What is more, the reader should remember that using traditional methods like BEPU typically demands also a great effort of preliminary sensitivity analysis, which does not have to be systematic. The E-BEPU provides a more systematic approach to studying different sequences, and it is based on probabilistic methods. Analysts, designers, and regulatory bodies will gain more knowledge, both qualitative and quantitative, about nuclear installation safety, which can be considered an advantage.

The reader should be aware that this study has important limitations. The main intention was to apply and discuss with practical examples a new methodology for the first time, and it should be considered the main contribution and novelty of this work. The presented study is the first step in the process of methodology testing, and a great effort is necessary to test it further. There are also other specific limitations; the applied NPP is generic and is based on generic input data. The applied TH models and also PSA models have limitations inherent to the generic approach, and the level of details in comparison to real plants is reduced. What is more, selected uncertainty parameters can be reconsidered and studied more deeply for the SL-LOCA. Moreover, only the short-term response was studied with some specific assumptions for initial/boundary conditions. It can be argued that more detailed methods could be more suitable, for example, to cover the 3D neutron kinetics and power distributions with proper core simulators. In general, in the safety analysis, details are important, and differences can easily propagate into not fully expected outcomes. Hence, the reader should be careful in drawing too far-reaching conclusions about transient when using generic models. What is more, any modern methodology should also include acceptability tests for each step of the activity and the qualification of applied computational models [60]. For presented exercise, it was performed with engineering judgment, qualitative comparisons, and detailed checks, but they should be added as a more formal part of the methodology.

In this paper, only the 'left' successful branch of the methodology was demonstrated, and it became clear that more research effort is necessary to fully reveal the power of this approach and test it with other applications. In further research activities, the right branch of the algorithm (Blocks #10-14 in Fig. 1) has to be explicitly tested with some properly designed test case (e.g., power uprate) to clearly show methodology performance. What is more, further activities should cover the application of the methodology for DEC cases and also lower DBC classes.

This work presents an attempt to analyze Design Basis Accident with E-BEPU. Due to the general importance of the more conservative methods in the design basis domain, which have a long history and strong foundation in the Nuclear Industry, it is unlikely that combined methods like E-BEPU will replace the current best approach (e.g., BEPU) in the near future. The risk-informed combined probabilistic-deterministic approaches, like the proposed E-BEPU, are more promising for 'real' applications in the near term for other plant state categories, AOOs and DECs.

From the perspective of current IAEA DSA standards, Option-4 is dedicated to studying AOO or DEC, and BEPU (Option-3) methods are dedicated (also) to the DBA analysis. The methodology was designed with an intention to allow performing the design verification for the full scope of possible occurrences, including DBAs. Nevertheless, the discussion in Section 1 indicated that it could not be classified as an Option-4 defined in the IAEA SSG-2 Revision-1. In such a situation, it can be classified as an alternative approach to DSA. However, when used for non-DBA applications, this methodology can be considered a possible realization of Option-4 and an alternative approach to DSA analysis.

The presented E-BEPU is a design verification methodology and, in principle, can be applied not only in Nuclear Industry but can be used to check any design of complex industrial installation which is dependent on the strict regulations.

CRediT authorship contribution statement

Piotr Mazgaj: Software, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization, Funding acquisition. **Piotr Darnowski:** Software, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Aleksej Kaszko:** Software, Methodology, Validation, Formal analysis, Investigation, Writing – review & editing. **Javier Hortal:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing, Supervision. **Milorad Dusic:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision. **Rafael Mendizábal:** Conceptualization, Methodology. **Fernando Pelayo:** Conceptualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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